DAY12-LINUX-NETWORK

THEORY (answer all 20 questions in your own words):

1. What is a network interface?
   A network interface lets the Linux system communicate with a network.
   Eg: etho0, ens33, wlan0, wlp2s0.

2. What is the loopback interface?
   Linux command has lo the loopback interface, with IPv4 127.0.0.1
   Identification: Interface name, Ipv4, Ipv6.

3. What does ip addr display:
   Display Network interface, IP address and subnet infos.

4. What is the purpose of ip link?
   It displays and manage network interface by enabling or disabling them.

5. What does hostname -I display?
   Display Ip address to system.

6. What does ip route show?
   Display system routing table.

7. What is a default route?
   A default route is the route used when no specific route matches the destination IP address.

8. What does ping test?
   Used mainly for checking internet connectivity ICMP echo request and checking for replies.

9. Why can a failed ping be misleading?
   A firewall may block ICMP packets even when the destination is reachable through other services.

10. What is DNS?
    DNS (Domain Name System) translates domain names into IP addresses.

11. What does getent hosts do?
    Looks up host information, such as IP addresses, using the system's configured name-resolution sources.

12. What information does ip neigh display?
    Displays neighbouring devices' IP addresses, link-layer (MAC) addresses, and reachability states.

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13. What is a socket?
    A socket is an endpoint used by applications to communicate over a network.

14. What does ss -tuln display?
    Displays listening TCP and UDP sockets with numerical IP addresses and port numbers.

15. What is the difference between TCP and UDP conceptually?
    TCP is connection-oriented and provides reliable, ordered delivery. UDP is connectionless and does not guarantee delivery or order.

16. What is curl used for?
    Used to transfer data to or from servers using protocols such as HTTP and HTTPS.

17. What does curl -I request?
    Requests only the HTTP response headers from a server, without downloading the response body.

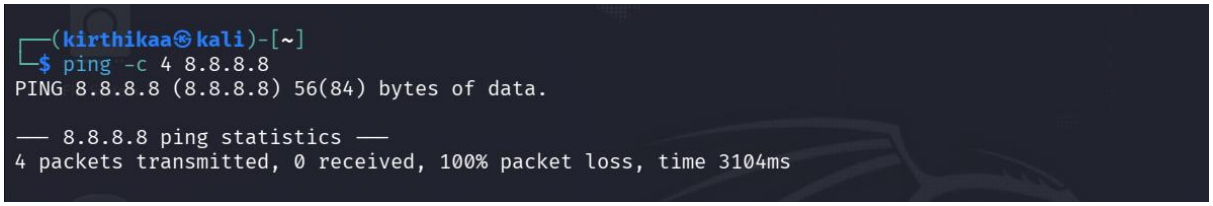
18. What is traceroute used for?
    Shows the network hops packets take to reach a destination and helps troubleshoot routing problems.

19. Why is ss commonly preferred over netstat on modern Linux?
    ss is generally faster and provides detailed socket information. It is also the modern replacement commonly recommended for netstat.

20. Why are Linux networking commands useful in cybersecurity?
    They help analysts inspect network configurations, identify listening ports, troubleshoot connectivity, and investigate suspicious connections.

<u>TROUBLESHOOTING QUESTION (from lab Task 10):</u>

A Linux system can ping 8.8.8.8, but ping google.com fails because the hostname cannot be resolved. What type of problem is most likely, and which command from today would you use first to confirm it?



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DAY 12 PRACTICAL LAB — Linux Network Investigation Lab

AUTHORIZED ENVIRONMENT ONLY: your own Kali VM / Linux VM / Jayvik Labs lab. No scanning required — and none allowed.

TASK 1 — ip addr: record Loopback interface, Primary interface, IPv4 address, Prefix.

```
┌──(kirthikaa㉿kali)-[~]
└─$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:af:ed:06 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86198sec preferred_lft 86198sec
    inet6 fd17:625c:f037:2:c188:342c:972f:77d6/64 scope global temporary dynamic
       valid_lft 86199sec preferred_lft 14199sec
    inet6 fd17:625c:f037:2:a00:27ff:feaf:ed06/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86199sec preferred_lft 14199sec
    inet6 fe80::a00:27ff:feaf:ed06/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

TASK 2 — ip link: record Interface state and MAC address.

```
┌──(kirthikaa㉿kali)-[~]
└─$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:af:ed:06 brd ff:ff:ff:ff:ff:ff
```

TASK 3 — ip route: record Default gateway, Primary interface, Default route line.

```
┌──(kirthikaa㉿kali)-[~]
└─$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
```

TASK 4 — ping -c 4 127.0.0.1: Did loopback respond (Yes/No)? What does this test? (Your local TCP/IP stack.)

```
┌──(kirthikaa㉿kali)-[~]
└─$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.046 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.042 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.070 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.037 ms

── 127.0.0.1 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3060ms
rtt min/avg/max/mdev = 0.037/0.048/0.070/0.012 ms
```

TASK 5 — ping -c 4 google.com: record packets transmitted / received / loss. Results differ per network — that's normal.

```
┌──(kirthikaa㉿kali)-[~]
└─$ ping -c 4 google.com
PING google.com (172.217.25.78) 56(84) bytes of data.

── google.com ping statistics ──
4 packets transmitted, 0 received, 100% packet loss, time 3058ms
```

TASK 6 — getent hosts google.com: record a resolved IP (multiple IPs is fine).

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```
┌──(kirthikaa㉿kali)-[~]
└─$ getent hosts google.com
2404:6800:4001:c01::8b google.com
2404:6800:4001:c01::8a google.com
2404:6800:4001:c01::65 google.com
2404:6800:4001:c01::71 google.com
```

TASK 7 — ip neigh: record ONE neighbour entry (IP/MAC/state) IF your system already has one. If none appears, leave it — do NOT generate traffic at other devices to force one.

```
┌──(kirthikaa㉿kali)-[~]
└─$ ip neigh
10.0.2.2 dev eth0 lladdr 52:54:00:12:35:00 STALE
fd17:625c:f037:2::2 dev eth0 lladdr 52:54:00:12:35:00 router STALE
fe80::2 dev eth0 lladdr 52:54:00:12:35:00 router STALE
```

TASK 8 — ss -tuln: record up to three listening ports if present (protocol / local address / port). No specific ports are required.

```
┌──(kirthikaa㉿kali)-[~]
└─$ ss -tuln
Netid          State           Recv-Q          Send-Q                    Local Address:Port                    Peer Address:Port
```

TASK 9 — curl -I https://example.com: record HTTP status, Content-Type, any other header.

```
┌──(kirthikaa㉿kali)-[~]
└─$ curl -I https://google.com
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-k7_R2eRDqxQmh8dCDqDgig' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
date: Fri, 16 Oct 2026 09:48:25 GMT
expires: Sun, 15 Nov 2026 09:48:25 GMT
cache-control: public, max-age=2592000
server: gws
content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
```

TASK 10 — Troubleshooting challenge: A Linux system can ping 8.8.8.8, but ping google.com fails because the hostname cannot be resolved. What type of problem is most likely? Write your answer in the assignment.

The most likely problem is a DNS resolution issue. The system has internet connectivity, but it cannot translate the hostname google.com into an IP address.